

Analytic Observables Along Recursive Complex Root Orbits

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Abstract

We study a simple source of recursively generated complex parameters and the analytic objects obtained by observing classical functions along its orbit. Let

$$T_r(z) = \text{Root}_r^{\text{pr}}(-z), \quad r \geq 2,$$

where the principal branch is fixed by $\text{Arg } z \in (-\pi, \pi]$. The familiar square-root recursion is the case $r = 2$. We first solve the principal dynamics explicitly. In the square-root case the orbit admits a closed formula and converges to an attracting two-cycle rather than to a fixed point. More generally, the principal map T_r has the explicit two-cycle $e^{\pm i\pi/(r+1)}$.

We then separate two structures that are often conflated: the orbit of a chosen branch and the full family of roots. The latter gives a nested hierarchy of roots of unity, a direct-limit phase group, and an inverse-limit space of coherent root histories. These constructions provide a natural bridge between complex phase, symbolic branch addresses, and finite-resolution representations.

Finally, we formulate an elementary sampling principle: if A is holomorphic near the attracting cycle, then $A(T_r^n(z_0))$ has an explicit two-subsequence asymptotic expansion. Classical Laplace and Mellin transforms, and a repaired zeta example, are then understood as analytic observables sampled along the root orbit rather than as newly invented transforms. The resulting framework is deliberately conservative: the underlying complex-analytic and algebraic objects are classical, while the purpose of the paper is to place branch-sensitive recursive root dynamics and analytic observation within one reusable mathematical structure.

1 Introduction

Complex roots are multivalued. Once a branch is fixed, a root operation becomes a single-valued map on its chosen domain; if all roots are retained, the same algebraic equation instead defines a correspondence with several inverse branches. These two viewpoints lead to different mathematical objects and should not be identified.

The starting point of this paper is the principal recursion

$$z_{n+1} = T_r(z_n) := \text{Root}_r^{\text{pr}}(-z_n), \quad r \geq 2. \quad (1.1)$$

The case $r = 2$, beginning with $z_1 = i$, arose in earlier exploratory work on recursively defined “higher-order imaginary numbers.” Those experiments also motivated proposed extensions of Fourier and Mellin analysis and a deformation of the Riemann zeta function. The present paper keeps the useful mathematical question while removing claims that depended on treating ordinary complex numbers as a new number system or on replacing a classical transform by a renamed copy of itself.

There is substantial prior work on iterated complex radicals. The convergence problem for fixed determinations of the square root predates the present discussion and was studied by Ogilvy, Gerber, Rohde, and Walker [1, 2, 3, 4]. The purpose here is therefore not to claim that

recursive complex radicals are new. Instead, we isolate a structural viewpoint that can be reused across analysis, dynamics, symbolic representation, and computation.

The guiding principle is

$$\boxed{\text{recursive root dynamics} \longrightarrow \text{analytic observables sampled along the orbit.}} \quad (1.2)$$

Thus a transform or special function need not itself be altered. It can be evaluated along a parameter trajectory generated by the root dynamics. This distinction is elementary, but it prevents several conceptual and technical errors and gives a common language to otherwise unrelated constructions.

2 Principal branch and the recursive root map

Throughout, for $z \neq 0$, let

$$\text{Arg } z \in (-\pi, \pi], \quad \text{Log } z = \log |z| + i \text{Arg } z.$$

For $r \geq 2$, define

$$\text{Root}_r^{\text{Pr}}(z) = \exp\left(\frac{1}{r} \text{Log } z\right), \quad z \neq 0, \quad (2.1)$$

and set $\text{Root}_r^{\text{Pr}}(0) = 0$.

Definition 2.1 (Principal negative-root map). For $r \geq 2$, define

$$T_r(z) = \text{Root}_r^{\text{Pr}}(-z).$$

On the unit circle, if $z = e^{i\theta}$ with $\theta \in (-\pi, \pi]$, then

$$T_r(e^{i\theta}) = \exp\left(\frac{i}{r} \text{PV}(\theta + \pi)\right), \quad (2.2)$$

where PV denotes reduction to $(-\pi, \pi]$.

The branch convention is essential. The unreduced affine expression $(\theta + \pi)/r$ is a valid lift of the argument map, but it is not itself the principal argument after every iteration.

2.1 The square-root orbit

The square-root case admits a particularly simple formula.

Theorem 2.2 (Exact square-root orbit). *Let*

$$z_1 = i, \quad z_{n+1} = T_2(z_n),$$

and write $z_n = e^{i\theta_n}$ with $\theta_n \in (-\pi, \pi]$. Then

$$\boxed{\theta_n = (-1)^{n+1} \frac{\pi}{3} + \frac{\pi}{3 \cdot 2^n}} \quad (2.3)$$

and therefore

$$\boxed{\lim_{m \rightarrow \infty} z_{2m+1} = e^{i\pi/3}, \quad \lim_{m \rightarrow \infty} z_{2m} = e^{-i\pi/3}.} \quad (2.4)$$

Proof. The initial value is $\theta_1 = \pi/2$, which agrees with (2.3). Suppose first that n is odd and (2.3) holds. Then

$$\theta_n + \pi = \frac{4\pi}{3} + \frac{\pi}{3 \cdot 2^n},$$

which lies above π , so principal-value reduction subtracts 2π . Hence

$$\theta_{n+1} = \frac{1}{2} \left(-\frac{2\pi}{3} + \frac{\pi}{3 \cdot 2^n} \right) = -\frac{\pi}{3} + \frac{\pi}{3 \cdot 2^{n+1}}.$$

If n is even, then

$$\theta_n + \pi = \frac{2\pi}{3} + \frac{\pi}{3 \cdot 2^n},$$

which already lies in $(-\pi, \pi]$. Thus

$$\theta_{n+1} = \frac{1}{2} \left(\frac{2\pi}{3} + \frac{\pi}{3 \cdot 2^n} \right) = \frac{\pi}{3} + \frac{\pi}{3 \cdot 2^{n+1}}.$$

This proves the formula by induction. Taking even and odd subsequences gives (2.4). $\square$ $\square$

Remark 2.3. The sequence therefore does not converge to 1. The correct limiting object is an attracting two-cycle. The earlier experimental sequence

$$i, e^{-i\pi/4}, e^{3i\pi/8}, e^{-5i\pi/16}, \dots$$

is consistent with (2.3). The distinction between convergence to a fixed point and convergence to a periodic orbit is fundamental for all later constructions.

2.2 The general root-order family

The same phenomenon persists for every $r \geq 2$.

Proposition 2.4 (Explicit attracting two-cycle). *For every $r \geq 2$, the principal map T_r has the two-cycle*

$$q_+^{(r)} = e^{i\pi/(r+1)}, \quad q_-^{(r)} = e^{-i\pi/(r+1)}, \quad (2.5)$$

so that

$$T_r(q_+^{(r)}) = q_-^{(r)}, \quad T_r(q_-^{(r)}) = q_+^{(r)}.$$

Moreover, whenever the orbit remains away from the branch endpoint, the two-step phase map contracts by the factor r^{-2} .

Proof. For $q_+^{(r)}$, its argument is $\pi/(r+1)$. Since

$$\frac{\pi}{r+1} + \pi = \frac{r+2}{r+1}\pi > \pi,$$

principal reduction subtracts 2π , giving $-\pi/(r+1)$. Division by r yields $-\pi/(r+1)$. The calculation for $q_-^{(r)}$ is analogous:

$$-\frac{\pi}{r+1} + \pi = \frac{r}{r+1}\pi,$$

which is already principal, and division by r gives $\pi/(r+1)$. On any pair of continuity intervals for the branch map, each application divides angular differences by r ; two applications therefore divide them by r^2 . $\square$ $\square$

3 The full root correspondence

Fix $r \geq 2$. The principal map selects one root. The full equation

$$w^r = z$$

has r roots when $z \neq 0$. The natural finite phase sets are

$$\mu_{r^n} = \left\{ e^{2\pi i k / r^n} : 0 \leq k < r^n \right\}, \quad n \geq 0. \quad (3.1)$$

They form a nested family

$$\mu_1 \subset \mu_r \subset \mu_{r^2} \subset \cdots$$

Definition 3.1 (Resolution phase group). Define

$$G_r := \bigcup_{n \geq 0} \mu_{r^n}.$$

Proposition 3.2 (Direct-limit description). *There is a natural group isomorphism*

$$\boxed{G_r \cong \mathbb{Z}[1/r]/\mathbb{Z}.} \quad (3.2)$$

Proof. Map

$$\frac{k}{r^n} + \mathbb{Z} \mapsto e^{2\pi i k / r^n}.$$

Changing the representative by an integer does not change the exponential, addition in $\mathbb{Z}[1/r]/\mathbb{Z}$ corresponds to multiplication of roots of unity, and the kernel is trivial. Surjectivity is immediate from the definition of G_r . $\square$ $\square$

The second structure is obtained by reversing the maps.

Definition 3.3 (Coherent root histories). Let

$$q_n : \mu_{r^{n+1}} \rightarrow \mu_{r^n}, \quad q_n(z) = z^r.$$

Define

$$H_r := \varprojlim \left(\mu_1 \xleftarrow{q_0} \mu_r \xleftarrow{q_1} \mu_{r^2} \xleftarrow{q_2} \cdots \right). \quad (3.3)$$

An element of H_r is a sequence $(z_n)_{n \geq 0}$ satisfying $z_{n+1}^r = z_n$.

Choose compatible primitive roots $\xi_n = e^{2\pi i / r^n}$ satisfying $\xi_{n+1}^r = \xi_n$. If $z_n = \xi_n^{a_n}$, then coherence is exactly

$$a_{n+1} \equiv a_n \pmod{r^n}.$$

Hence

$$\boxed{H_r \cong \varprojlim_n \mathbb{Z}/r^n \mathbb{Z}.} \quad (3.4)$$

For prime $r = p$, the right-hand side is the ring of p -adic integers $\mathbb{Z}_p$; for general r , the Chinese remainder theorem gives the product of the corresponding p -adic integer rings over the primes dividing r .

Proposition 3.4 (No homomorphic branch selector). *For $n \geq 1$, the exact sequence*

$$1 \longrightarrow \mu_r \longrightarrow \mu_{r^{n+1}} \xrightarrow{z \mapsto z^r} \mu_{r^n} \longrightarrow 1 \quad (3.5)$$

has no group-homomorphic section.

Proof. Suppose $s : \mu_{r^n} \rightarrow \mu_{r^{n+1}}$ were a homomorphism satisfying $s(z)^r = z$. If z generates μ_{r^n} , then $s(z)$ has order r^{n+1} . But the order of every element in the image of a homomorphism from a group of order r^n divides r^n . This is a contradiction. $\square$ $\square$

Remark 3.5. This obstruction gives a precise mathematical meaning to the statement that branch selection adds structure. A principal root, a measurable selector, a symbolic rule, or a random selector may exist, but none is a group-theoretic inverse to the power map.

4 Branch addresses and finite-resolution phase

At level n , write a root as

$$z_n = c_n \exp\left(\frac{2\pi i a_n}{r^n}\right), \quad a_n \in \mathbb{Z}/r^n\mathbb{Z},$$

where (c_n) is any fixed reference root chain satisfying the same recurrence. The compatibility relation gives

$$a_{n+1} \equiv a_n \pmod{r^n}.$$

Equivalently, there are digits $d_j \in \{0, 1, \dots, r-1\}$ such that

$$a_n = d_0 + d_1 r + \dots + d_{n-1} r^{n-1}. \quad (4.1)$$

Thus an infinite branch is naturally encoded by a one-sided digit sequence

$$(d_0, d_1, d_2, \dots),$$

with the associated coherent residue class in H_r .

The finite phase readout of an address $a \in H_r$ is

$$R_n(a) = \exp\left(\frac{2\pi i a_n}{r^n}\right). \quad (4.2)$$

It satisfies

$$R_{n+1}(a)^r = R_n(a).$$

Proposition 4.1 (First distinguishing digit). *Suppose $a, b \in H_r$ agree modulo r^n but differ modulo r^{n+1} . Then for some $c \in \{1, \dots, r-1\}$,*

$$a_{n+1} - b_{n+1} \equiv cr^n \pmod{r^{n+1}},$$

and

$$|R_{n+1}(a) - R_{n+1}(b)| = 2 \sin\left(\frac{\pi c}{r}\right) \geq 2 \sin\left(\frac{\pi}{r}\right). \quad (4.3)$$

Proof. The quotient of the two phase readouts is

$$\frac{R_{n+1}(a)}{R_{n+1}(b)} = e^{2\pi i c/r}.$$

The chord-length formula on the unit circle gives the result. □ □

This makes the resolution hierarchy quantitative: the first branch digit at which two coherent histories differ produces a definite phase separation at that resolution.

5 Analytic observables along the orbit

We now state the central analytic principle of the paper.

Let T_2 be the square-root map and let (z_n) be an orbit with $z_1 = i$. Put

$$q_+ = e^{i\pi/3}, \quad q_- = e^{-i\pi/3}.$$

Theorem 5.1 (Orbit-observable asymptotics). *Let A be holomorphic on neighborhoods of q_+ and q_- . Then, as $m \rightarrow \infty$,*

$$A(z_{2m+1}) = A(q_+) + \frac{i\pi q_+}{3 \cdot 2^{2m+1}} A'(q_+) + O(4^{-2m}), \quad (5.1)$$

$$A(z_{2m}) = A(q_-) + \frac{i\pi q_-}{3 \cdot 2^{2m}} A'(q_-) + O(4^{-2m}). \quad (5.2)$$

More generally, the same statement holds for any orbit converging to the two-cycle with the corresponding first-order orbit error.

Proof. By theorem 2.2,

$$z_n = q_{\sigma_n} \exp\left(i \frac{\pi}{3 \cdot 2^n}\right),$$

where $\sigma_n = +$ for odd n and $\sigma_n = -$ for even n . Hence

$$z_n - q_{\sigma_n} = i q_{\sigma_n} \frac{\pi}{3 \cdot 2^n} + O(4^{-n}).$$

Taylor's theorem for the holomorphic function A at q_{σ_n} gives

$$A(z_n) = A(q_{\sigma_n}) + A'(q_{\sigma_n})(z_n - q_{\sigma_n}) + O(|z_n - q_{\sigma_n}|^2),$$

which yields the asserted formulas. □ □

The theorem is deliberately general. The function A may be a transform, a special function, a spectral multiplier, a response function, or any other analytic observable defined near the two limiting parameters.

5.1 Laplace sampling

Let $f : [0, \infty) \rightarrow \mathbb{C}$ have a Laplace transform in a domain containing the relevant points. The natural recursively sampled family is

$$L_n(\omega) := \mathcal{L}f(z_n \omega), \quad (5.3)$$

where $\mathcal{L}f(s) = \int_0^\infty f(t) e^{-st} dt$.

This is not a new Laplace transform. It is the classical Laplace transform evaluated along the root orbit. Whenever $s \mapsto \mathcal{L}f(s)$ is holomorphic near $q_\pm \omega$, theorem 5.1 applies directly.

This reformulation also fixes a problem in the earlier exploratory Fourier construction: an example based on $f(t) = e^{-2t}$ over the whole real line cannot be used as an ordinary Fourier integral because the function is not integrable as $t \rightarrow -\infty$. The correct procedure is to specify the domain and the transform hypotheses before making the recursive parameter substitution.

5.2 Mellin sampling

For a Mellin-transformable function f ,

$$\mathcal{M}f(s) = \int_0^\infty f(t) t^{s-1} dt,$$

the natural sampled family is

$$M_n(s) := \mathcal{M}f(s + z_n - 1). \quad (5.4)$$

The expression sometimes called an “imaginary Mellin transform” in earlier work,

$$\int_0^\infty f(t) t^{s+J-2} dt,$$

is exactly

$$\mathcal{M}f(s + J - 1). \quad (5.5)$$

Thus its mathematical content is not the creation of a new integral transform; the recursive information lies in the trajectory $J = z_n$ through Mellin parameter space.

6 A repaired zeta example

The same distinction clarifies the earlier zeta experiment. Let

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s) \quad (6.1)$$

be the completed Riemann zeta function. It is entire and satisfies

$$\xi(s) = \xi(1-s).$$

For any fixed $J \in \mathbb{C}$, define the shifted observable

$$\Xi_J(s) := \xi(s+J-1). \quad (6.2)$$

This is the analytically clean form of the deformation that is useful for the present framework.

Theorem 6.1 (Translated zeta symmetry). *For every fixed $J \in \mathbb{C}$,*

$$\boxed{\Xi_J(s) = \Xi_J(2-J-s)}. \quad (6.3)$$

If ρ is a zero of ξ , then

$$\boxed{s = \rho + 1 - J} \quad (6.4)$$

is a zero of Ξ_J . Conversely, every zero of Ξ_J arises in this way.

Proof. By the functional equation of ξ ,

$$\Xi_J(2-J-s) = \xi(2-J-s+J-1) = \xi(1-(s+J-1)) = \xi(s+J-1) = \Xi_J(s).$$

The zero statement follows from the definition. □

Remark 6.2. The symmetry center in the s -plane is therefore

$$1 - \frac{J}{2},$$

not $1/2$ unless $J = 1$. Thus the appropriate conclusion is not that the zeta functional equation has mysteriously disappeared, but that the coordinate system has been translated. The earlier deformed-zeta calculation was useful as an experiment precisely because it exposed this distinction.

Now let $J = z_n$ be the recursive square-root orbit. The corresponding symmetry centers are

$$c_n = 1 - \frac{z_n}{2}, \quad (6.5)$$

and hence

$$c_{2m+1} \rightarrow 1 - \frac{1}{2}e^{i\pi/3}, \quad c_{2m} \rightarrow 1 - \frac{1}{2}e^{-i\pi/3}. \quad (6.6)$$

Thus the recursive root dynamics induces an explicit two-cycle of symmetry centers for this family of analytically shifted zeta observables.

7 Finite phase resolution and multiresolution analysis

The root tower also supplies a natural finite-resolution phase hierarchy. Let

$$V_n = \{f : H_r \rightarrow \mathbb{C} : f \text{ depends only on } a \bmod r^n\}.$$

Then

$$\dim V_n = r^n, \quad V_n \subset V_{n+1}.$$

The characters of the finite quotient $\mathbb{Z}/r^n\mathbb{Z}$ are

$$\chi_{n,k}(a) = e^{2\pi i k a / r^n}, \quad 0 \leq k < r^n,$$

and form a basis for V_n .

Define

$$W_n = V_{n+1} \ominus V_n$$

with respect to the normalized counting inner product on finite quotients and the induced projective limit measure. Then

$$\dim W_n = (r-1)r^n. \quad (7.1)$$

The corresponding new characters are exactly those whose frequencies occur at level r^{n+1} but not at level r^n .

Proposition 7.1 (Radix refinement law). *Let $f_n : \mathbb{Z}/r^n\mathbb{Z} \rightarrow \mathbb{C}$, and define its refinement by*

$$(J_n f_n)(k + jr^n) = f_n(k), \quad 0 \leq k < r^n, \quad 0 \leq j < r.$$

For the normalized discrete Fourier transform,

$$\widehat{J_n f_{n+1}}(m) = \begin{cases} \widehat{f_n}(m/r), & r \mid m, \\ 0, & r \nmid m. \end{cases} \quad (7.2)$$

Proof. Substituting the refinement definition gives

$$\widehat{J_n f_{n+1}}(m) = \frac{1}{r^{n+1}} \left(\sum_{k=0}^{r^n-1} f_n(k) e^{-2\pi i m k / r^{n+1}} \right) \left(\sum_{j=0}^{r-1} e^{-2\pi i m j / r} \right).$$

The second factor is r when $r \mid m$ and is zero otherwise. When $m = r\ell$, the remaining expression is $\widehat{f_n}(\ell)$. □

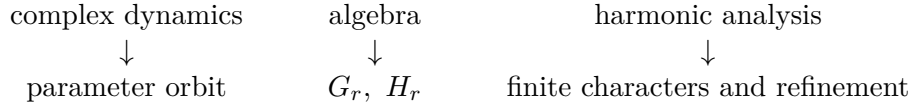
This provides a precise multiresolution interpretation. Coarse information is carried forward by frequencies divisible by r , while the remaining frequencies represent detail that appears only when the phase resolution is refined.

8 What this framework does and does not claim

The constructions in this paper use standard complex analysis, cyclic groups, inverse limits, discrete Fourier analysis, and classical zeta theory. No new field of numbers is introduced merely by naming recursively generated complex values. Likewise, the sampled Laplace and Mellin families are not new integral transforms in themselves.

The intended contribution is structural. A branch-sensitive recursive root map supplies a deterministic parameter orbit; the complete root correspondence supplies a hierarchy of finite

phase spaces and coherent branch histories; and analytic functions can be observed along those structures. The same framework can therefore be read from several directions:



while transforms and special functions enter as observables rather than as separate ad hoc theories.

Several questions remain open. Among them are the operator theory of branch selectors on H_r , the spectral theory of transfer operators associated with phase refinement, the relation between deterministic principal dynamics and random branch processes, and the extension of the analytic-observable principle from the square-root map to general r -root maps and other branch-generated complex parameter systems.

9 Conclusion

A recursive complex root is an elementary operation, but once branch selection, full inverse structure, and analytic observation are kept separate, it produces a useful hierarchy of mathematical objects.

For the principal square-root orbit beginning at i , the exact formula

$$\theta_n = (-1)^{n+1} \frac{\pi}{3} + \frac{\pi}{3 \cdot 2^n}$$

shows that the asymptotic object is an attracting two-cycle. For general root order r , the cycle becomes

$$\left\{ e^{i\pi/(r+1)}, e^{-i\pi/(r+1)} \right\}.$$

Retaining all roots produces the phase group

$$G_r = \bigcup_n \mu_{r^n}$$

and the coherent-history space H_r , while finite quotients provide exact phase resolutions and Fourier refinement laws.

Most importantly, if an analytic object A is sampled along the recursive orbit, its asymptotic behavior is controlled by the attracting cycle and the local Taylor expansion of A . This principle turns several earlier exploratory constructions into instances of one question:

What information is revealed when a classical analytic object is observed
along a branch-generated complex orbit?

That question is broad enough to invite applications, but precise enough to admit proofs.

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